AIWSEN: Adaptive Information Weighting and Synchronized Enhancement Network for Hyperspectral Change Detection

Lanxin Wu, Jiangtao Peng, *Senior Member, IEEE*, Bing Yang, Weiwei Sun, *Senior Member, IEEE*, and Zhijing Ye

***Abstract*— Hyperspectral image (HSI) change detection (CD) plays a crucial role in remote sensing observation. It leverages the abundant spectral and spatial information in bi-temporal HSIs to identify subtle Earth surface changes. Most current deep-learning-based HSI CD methods primarily utilize convolutional neural networks or transformers to extract features from bi-temporal images. However, these methods lack an effective attention mechanism to enhance differential features. In addition, they do not fully leverage the aggregation relationship between the features of bi-temporal images to extract interaction features. To address these challenges, we propose a novel adaptive information weighting and synchronized enhancement network (AIWSEN) for HSI CD. This network employs the information entropy to capture change features specific to the CD task and enhances bi-temporal interaction features. Specifically, an adaptive information weighting attention module (AIWAM) leverages the maximum discrete entropy theorem to capture the difference information. A dual-time synchronic change enhancing module (DSCEM) is designed to extract features by interactively aggregating features from bi-temporal HSIs to enhance difference features. A bi-temporal image feature selection and fusion module (BFSFM) is constructed to filter out important features using forget and update gates. Experimental results on three HSI CD datasets demonstrate that the proposed AIWSEN method outperforms several state-of-the-art methods. The source code of the proposed AIWSEN will be released at https://github.com/creativeXin/AIWSEN.**

***Index Terms*— Adaptive information weighting, change detection (CD), hyperspectral image (HSI), synchronic enhancing.**

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I. Introduction

CHANGE detection (CD) is a crucial technology for monitoring the Earth's surface, with widespread applications in forest conservation, agricultural monitoring, and land cover [1], [2], [3]. CD aims to identify changes that have occurred in the same area over different periods [4]. Due to their abundant spectral information, hyperspectral images (HSIs) have been extensively used in various applications [5], [6] and have become a hot topic in the CD.

During the early years, traditional methods such as change vector analysis (CVA) [7], iteratively reweighted multivariate alteration detection (IR-MAD) [8], and support vector machine (SVM) [9] were commonly used. Although these methods made some progress in HSI change detection (HSI-CD), they tend to ignore the abundant spatial and spectral information in HSIs and rely on manually set thresholds.

Recently, convolutional neural networks (CNNs) have shown significant advantages in CD due to their ability to automatically extract relevant features and capture spatial information [10], [11]. Wang et al. [12] proposed a 2-D CNN to obtain high-level discriminative features and improve the generalization ability of CD. Ou et al. [13] introduced a CNN architecture with fast and slow bands to extract a discriminative feature group. Huang et al. [14] employed a joint 2-D–3-D convolutional approach to simultaneously extract spatial and spectral features. Shi et al. [15] combined the CNN with a long short-term memory (LSTM) module to extract multiscale temporal–spatial–spectral features. Wu et al. [16] proposed a fully convolutional CD framework to unify unsupervised, weakly supervised, regional supervised, and fully supervised CD tasks into one end-to-end framework. Mou et al. [17] proposed the recurrent convolutional network architecture to extract multitemporal remote sensing image features. Lin et al. [18] proposed an end-to-end bilinear CNN to learn and utilize bi-temporal image relationships. Qu et al. [19] developed a multilevel encoder–decoder network to learn spatial and spectral features. Luo et al. [20] employed a CNN to obtain different receptive fields, achieving better results. Wang et al. [21] utilized a multilevel interactive architecture with CNN to capture multiscale and multilayer cross-temporal dependencies, balancing detection accuracy and complexity. Wu et al. [22] employed kernel principal component analysis convolution to extract representative fea-

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tures from multitemporal very high-resolution images. Zhang et al. [23] proposed a 3-D depthwise separable convolutional autoencoder to extract discriminative spatial and spectral features, achieving superior results. Despite the good performance of these CNN-based HSI CD methods, they still face challenges such as local receptive fields and a lack of long-range dependencies.

Transformer-based methods have gained significant attention in recent years due to their strong ability to capture long-term dependencies. For example, Chen et al. [24] employed transformer encoders and decoders to obtain semantic information for remote sensing CD. Shafique et al. [25] combined the advantages of CNN and transformers to achieve effective and efficient HSI CD. Hu et al. [26] introduced a multitemporal self-attention module to extract diverse feature maps. Zhang et al. [27] proposed a spectrum-aware transformer for feature fusion, resulting in improved performance. Tian et al. [28] developed a transformer branch and a CNN branch for adaptive contrastive learning. Ji et al. [29] combined CNN-transformer to facilitate effective coupling of feature information. However, it is worth noting that solely using transformers may not fully leverage the interactive information in bi-temporal images. To address this, Ding et al. [30] proposed a transformer module to mine the correlation and interaction in bi-temporal images. Sun et al. [31] introduced multihead cross attention to extract differential information. Song et al. [32] proposed a cross-transformer attention module to mine bi-temporal information. Li et al. [33] designed a spectral–temporal transformer to enrich discriminative features using the multihead self-attention mechanism. Xie et al. [34] proposed a cross-layer collaborative awareness module to enhance the transmission of information across layers. Wang et al. [35] developed a spectral–spatial–temporal transformer to extract spectral sequence information, spatial texture information, and temporal information. Ding et al. [36] converted HSI multiscale features into feature embeddings to aggregate temporal interactive information using transformer encoders. Wu et al. [37] proposed an enhanced difference attention mechanism to emphasize differential information. Chen et al. [38] combined CNN and transformer to extract HSI features. Although transformer-based methods have achieved superior results and performance, they still do not fully leverage the aggregation relationship between the features of bi-temporal images to enhance interactive features.

Many other attention-based methods are designed to focus on key information in images [39], [40], and they are widely used in the CD field. For instance, Dong et al. [41] proposed an abundance matrix learning module and a self-cross-hybrid attention network to learn hierarchical difference features. Ding and Li [42] utilized a convolutional block attention module to extract spatial and spectral features. Qu et al. [19] introduced a contextual information-guided attention module to fuse and exploit multilevel features. Wang et al. [43] used a joint spectral and spatial attention module to enhance spectral–spatial information simultaneously. Ding et al. [44] incorporated a channel and spatial attention module into the ResNet network block to improve feature extraction capabilities. Xu et al. [45] employed a channel attention module to achieve multiscale feature fusion in two parallel streams, resulting in better outcomes. Zhang et al. [46] proposed a transformer encoder to calculate self-attention matrices and obtain discriminative features. Luo et al. [47] proposed a multilevel res-attention block to obtain the difference information. However, most attention methods only enhance the spatial and channel information of HSIs, failing to utilize the distinctions between changed and unchanged pixels to extract and enhance differential features.

From the above research work, we can see significant potential in improving the extraction of exchange features from bi-temporal images and better-obtaining change information. First, CNN-based methods face difficulties in extracting long-range dependency information and are restricted by local receptive fields. Second, transformer-based methods cannot fully utilize the query, key, and value of bi-temporal images to obtain exchange information. Finally, CD methods using attentional mechanisms are adapted from other tasks, but they lack a specialized attention mechanism tailored for CD. To address these problems, we propose an adaptive information weighting and synchronized enhancement network (AIWSEN) for HSI CD. The proposed AIWSEN mainly consists of three modules: an adaptive information weighting attention module (AIWAM), a dual-time synchronic change enhancing module (DSCEM), and a bi-temporal image feature selection and fusion module (BFSFM). The AIWAM leverages the maximum discrete entropy theorem to capture the difference information of bi-temporal HSIs. The DSCEM employs an interactive aggregation method using features from bi-temporal HSIs, resulting in the extraction of interactive features while also enhancing difference features. The BFSFM is designed to fuse difference information with feature interaction information to improve CD results. The main contributions of this article are as follows.

1) A novel end-to-end network named AIWSEN is proposed for HSI CD. AIWSEN integrates AIWAM, DSCEM, and BFSFM, aiming to enhance change information, obtain exchange information, and fuse features to achieve superior results.
2) The AIWAM is proposed to leverage information entropy to enhance change information. Specifically designed for CD, it effectively distinguishes between areas with changed and unchanged pixels, ensuring accurate identification and highlighting of regions where changes have occurred.
3) The DSCEM is designed to extract interactive features from bi-temporal HSIs. It effectively utilizes bi-temporal HSIs for interactive feature aggregation, ensuring the extraction of interactive features while enhancing their difference information.

The rest of the article is organized as follows. Section II introduces the proposed AIWSEN. Section III presents the experimental results. Finally, Section IV concludes the article.

II. Methodology

The framework of the proposed AIWSEN is illustrated in Fig. 1. Given bi-temporal HSIs, the HSI patches are first extracted. Following this, the bi-temporal HSI patches are

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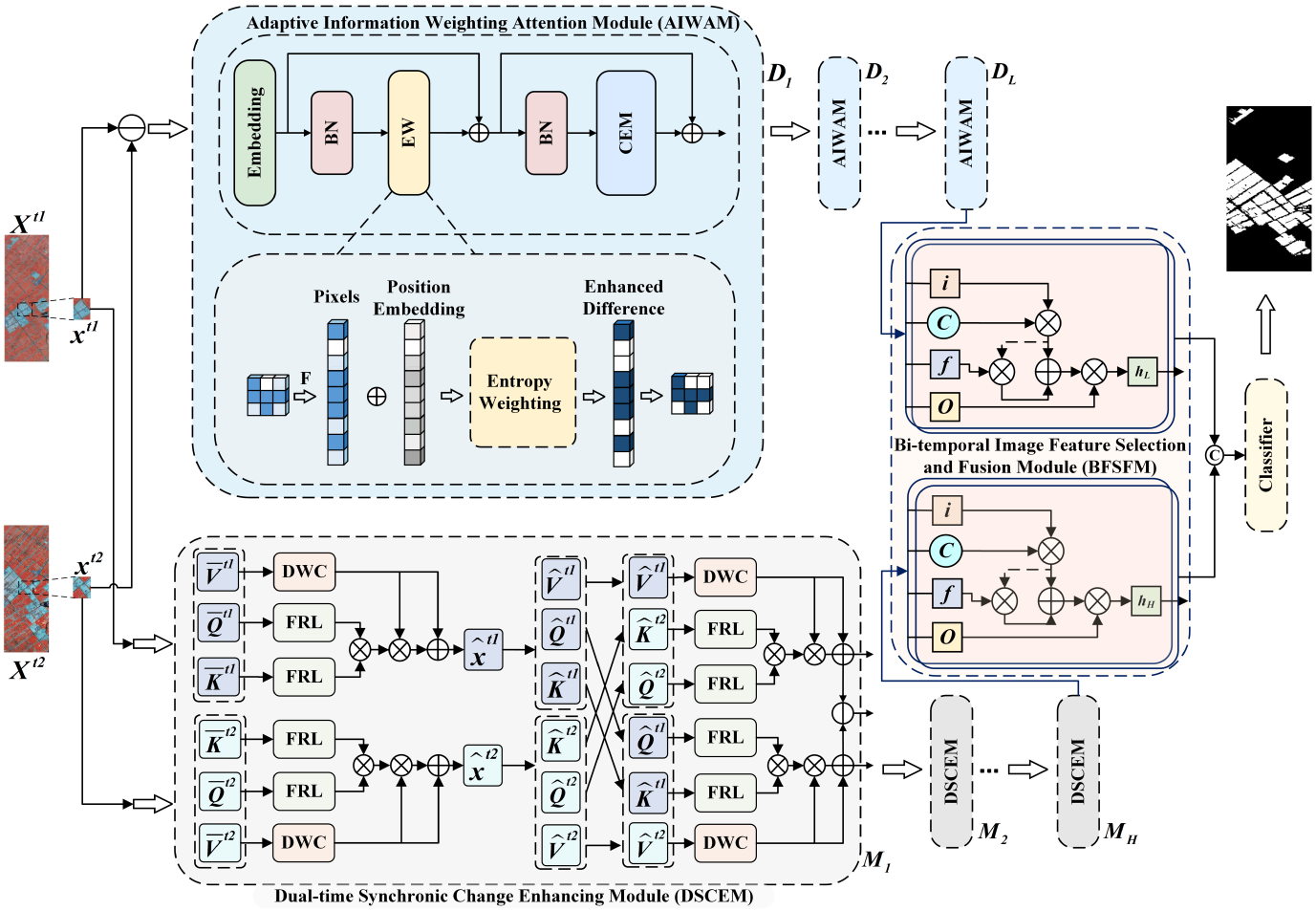


Fig. 1. Framework of the proposed AIWSEN for HSI CD includes AIWAM, DSCEM, and BFSFM. The AIWAM enhances the difference features, the DSCEM extracts the interactive features, and the BFSFM fuses the extracted features.

subtracted and then processed through a series of AIWAMs to extract difference features, which are enhanced using the proposed entropy weighting (EW). Simultaneously, the bi-temporal HSI patches are also processed through a series of DSCEMs for interactive feature extraction, aggregating different features for interaction. Next, the difference features and interactive features extracted by AIWAM and DSCEM are fused and input into the BFSFM. Finally, the fused features are fed into a classifier to produce a binary CD map.

Next sections, we provide a detailed description of each module, including AIWAM, DSCEM, BFSFM, and the loss function.

A. Adaptive Information Weighting Attention Module

Most existing attention-based HSI-CD methods use cross-attention or transformer strategies, which are often adapted from tasks like classification or object detection and lack a specific focus on CD. To address this, we design the AIWAM with EW as a core mechanism. EW highlights regions with significant spectral changes while suppressing areas with minimal variability, enabling the model to focus on the most informative regions.

Given an intermediate feature $\boldsymbol{D}$ as input of the AIWAM, we first pass it through an embedding layer to obtain features, and then apply a batch normalization (BN) layer to normalize the results. The specific process is as follows:

$$\boldsymbol{D}' = \mathrm{BN}(\boldsymbol{D}\boldsymbol{W}) \tag{1}$$

where $\boldsymbol{W}$ denotes the linear parameter. After normalization, we flatten the features and add relative position encoding to enhance the positional information

$$\bar{\boldsymbol{D}} = \mathrm{F}(\boldsymbol{D}') + \boldsymbol{E}_{\mathrm{pos}} \tag{2}$$

where $F(\cdot)$ denotes the flatten operation, and $\boldsymbol{E}_{\mathrm{pos}}$ denotes the relative position encoding. Subsequently, we represent the proportion of each pixel's value within the entire channel using a probability-based approach

$$\widetilde{\boldsymbol{D}} = \exp(\bar{\boldsymbol{D}}) / \sum \left(\exp(\bar{\boldsymbol{D}})\right) \tag{3}$$

where $\sum(\cdot)$ denotes the summation along the channel dimension.

Then, we send the $\widetilde{\boldsymbol{D}}$ into EW layer to obtain change attention $\bar{\boldsymbol{E}}$

$$\hat{\boldsymbol{W}} = -\sum \widetilde{\boldsymbol{D}} \ln(\widetilde{\boldsymbol{D}}) / \ln C \tag{4}$$

$$\bar{\boldsymbol{E}} = 1 - \hat{\boldsymbol{W}} \tag{5}$$

where $\sum$ denotes the summation along the channel dimension, and C denotes the number of channels.

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The EW layer focuses on obtaining change attention by emphasizing difference features and suppressing the unchanged features. If a pixel is a changing pixel, the fluctuation of pixel values across different bands (i.e., $\widetilde{\boldsymbol{D}}$) will be larger. According to the maximum discrete entropy theorem, the entropy value of the $\hat{\boldsymbol{W}}$ will be lower and the corresponding weight $\bar{\boldsymbol{E}}$ is larger. Conversely, if a pixel point is unchanged, the pixel value fluctuates less across different bands, resulting in a higher entropy value $\hat{\boldsymbol{W}}$ and a lower weight $\bar{\boldsymbol{E}}$. In this way, we obtain an attention map that focuses on changing pixels.

After that, the skip connection is performed to preserve the original input information and facilitate the learning of deeper features as

$$\boldsymbol{S} = \boldsymbol{D}\boldsymbol{W} + \bar{\boldsymbol{E}}. \tag{6}$$

Finally, we pass the feature $\boldsymbol{S}$ through a channel enhancement module (CEM) and generate the output of the AIWAM, which is summarized as follows:

$$\hat{\boldsymbol{D}} = \boldsymbol{S} + f_{\text{Conv}}^{1\times1}\Big(\text{G}\Big(f_{\text{Conv}}^{1\times1}(\text{BN}(\boldsymbol{S}))\Big)\Big) \tag{7}$$

where $f_{\text{Conv}}^{1\times1}$ denotes the 1×1 convolution kernel, and $\text{G}(\cdot)$ denotes the GELU activation function.

By utilizing AIWAM, we can generate a more effective attention map through a detailed process that involves calculating the probability distribution of each pixel and normalizing its entropy. This allows the model to more accurately focus on pixels that are undergoing changes, thereby significantly enhancing its ability to detect and identify regions of variation.

B. Dual-Time Synchronic Change Enhancing Module

To capture interaction characteristics between bi-temporal images, we propose the DSCEM, which combines a feature focus and synchronic enhancing architecture. Using a cross-attention mechanism, it aligns features to highlight changes while reducing noise in unchanged areas. Refinement layers further emphasize meaningful features, enhancing the model's ability to detect critical changes.

1) Feature Focus Architecture: Given input features $\boldsymbol{x}^{t1} \in \mathbb{R}^{s\times s\times C}$ and $\boldsymbol{x}^{t2} \in \mathbb{R}^{s\times s\times C}$ from bi-temporal HSIs, we first flatten the features into $\bar{\boldsymbol{x}}^{t1} \in \mathbb{R}^{s^2\times C}$ and $\bar{\boldsymbol{x}}^{t2} \in \mathbb{R}^{s^2\times C}$. Then, we derive the query, key, and value from $\bar{\boldsymbol{x}}^{t1}$ and $\bar{\boldsymbol{x}}^{t2}$, respectively,

$$\begin{cases} \bar{\boldsymbol{Q}}^{t1} = \bar{\boldsymbol{x}}^{t1}\bar{\boldsymbol{W}}_q^{t1} \\ \bar{\boldsymbol{K}}^{t1} = \bar{\boldsymbol{x}}^{t1}\bar{\boldsymbol{W}}_k^{t1} \\ \bar{\boldsymbol{V}}^{t1} = \bar{\boldsymbol{x}}^{t1}\bar{\boldsymbol{W}}_v^{t1} \end{cases} \tag{8}$$

$$\begin{cases} \bar{\boldsymbol{Q}}^{t2} = \bar{\boldsymbol{x}}^{t2}\bar{\boldsymbol{W}}_q^{t2} \\ \bar{\boldsymbol{K}}^{t2} = \bar{\boldsymbol{x}}^{t2}\bar{\boldsymbol{W}}_k^{t2} \\ \bar{\boldsymbol{V}}^{t2} = \bar{\boldsymbol{x}}^{t2}\bar{\boldsymbol{W}}_v^{t2} \end{cases} \tag{9}$$

where $\bar{\boldsymbol{W}}_q^{ti} \in \mathbb{R}^{C\times d}$, $\bar{\boldsymbol{W}}_k^{ti} \in \mathbb{R}^{C\times d}$, and $\bar{\boldsymbol{W}}_v^{ti} \in \mathbb{R}^{C\times d}$ are weight matrices for getting query, key, and value for t_i HSI ($i = 1, 2$), and d is the feature dimension.

Rather than simply using dot-product attention to compute the similarity score, we propose a method that enhances the similar features while suppressing dissimilar ones. Specifically, inspired by [48], we push each vector toward its nearest features to highlight similar features

$$\text{Sim}\Big(\bar{\boldsymbol{Q}}^{t1}, \bar{\boldsymbol{K}}^{t1}\Big) = \text{FRL}\Big(\bar{\boldsymbol{Q}}^{t1}\Big) \cdot \text{FRL}\Big(\bar{\boldsymbol{K}}^{t1}\Big)^T \tag{10}$$

where the function $\text{FRL}(\cdot)$ adjusts the direction of each query and key feature, bringing similar query–key pairs closer while pushing dissimilar pairs apart. The definition of $\text{FRL}(\cdot)$ is

$$\text{FRL}(\boldsymbol{x}) = \boldsymbol{f}_p\big(\text{ReLU}(\boldsymbol{x})\big), \qquad \boldsymbol{f}_p(\boldsymbol{x}) = \frac{\|\boldsymbol{x}\|}{\|\boldsymbol{x}^{**p}\|}\boldsymbol{x}^{**p} \tag{11}$$

where $\boldsymbol{x}^{**p}$ denotes element-wise power p of $\boldsymbol{x}$.

After obtaining the refined attention feature $\text{Sim}(\bar{\boldsymbol{Q}}^{t1}, \bar{\boldsymbol{K}}^{t1})$, we can extract $\hat{\boldsymbol{x}}^{t1}$ through the $\bar{\boldsymbol{V}}^{t1}$ value features

$$\hat{\boldsymbol{x}}^{t1} = \text{Sim}\Big(\bar{\boldsymbol{Q}}^{t1}, \bar{\boldsymbol{K}}^{t1}\Big)\bar{\boldsymbol{V}}^{t1} + \text{DWC}\Big(\bar{\boldsymbol{V}}^{t1}\Big) \tag{12}$$

where the depthwise convolution (DWC) is added to maintain feature diversity.

Similarly, we can obtain $\hat{\boldsymbol{x}}^{t2}$ as follows:

$$\hat{\boldsymbol{x}}^{t2} = \text{Sim}\Big(\bar{\boldsymbol{Q}}^{t2}, \bar{\boldsymbol{K}}^{t2}\Big)\bar{\boldsymbol{V}}^{t2} + \text{DWC}\Big(\bar{\boldsymbol{V}}^{t2}\Big). \tag{13}$$

2) Synchronic Enhancing Architecture: After that, we send $\hat{\boldsymbol{x}}^{t1}$ and $\hat{\boldsymbol{x}}^{t2}$ into the synchronic enhancing architecture. Here, we use a cross-query, key, and value approach to obtain the interaction features. Our motivation is to extract interactive features while simultaneously enhancing their difference information. The formulas are as follows:

$$\begin{cases} \hat{\boldsymbol{Q}}^{t1} = \hat{\boldsymbol{x}}^{t1}\hat{\boldsymbol{W}}_q^{t1} \\ \hat{\boldsymbol{K}}^{t1} = \hat{\boldsymbol{x}}^{t1}\hat{\boldsymbol{W}}_k^{t1} \\ \hat{\boldsymbol{V}}^{t1} = \hat{\boldsymbol{x}}^{t1}\hat{\boldsymbol{W}}_v^{t1} \end{cases} \tag{14}$$

$$\begin{cases} \hat{\boldsymbol{Q}}^{t2} = \hat{\boldsymbol{x}}^{t2}\hat{\boldsymbol{W}}_q^{t2} \\ \hat{\boldsymbol{K}}^{t2} = \hat{\boldsymbol{x}}^{t2}\hat{\boldsymbol{W}}_k^{t2} \\ \hat{\boldsymbol{V}}^{t2} = \hat{\boldsymbol{x}}^{t2}\hat{\boldsymbol{W}}_v^{t2} \end{cases} \tag{15}$$

$$\widetilde{\boldsymbol{x}}^{t1} = \text{Sim}\Big(\hat{\boldsymbol{Q}}^{t2}, \hat{\boldsymbol{K}}^{t2}\Big)\hat{\boldsymbol{V}}^{t1} + \text{DWC}\Big(\hat{\boldsymbol{V}}^{t1}\Big) \tag{16}$$

$$\widetilde{\boldsymbol{x}}^{t2} = \text{Sim}\Big(\hat{\boldsymbol{Q}}^{t1}, \hat{\boldsymbol{K}}^{t1}\Big)\hat{\boldsymbol{V}}^{t2} + \text{DWC}\Big(\hat{\boldsymbol{V}}^{t2}\Big) \tag{17}$$

where $\hat{\boldsymbol{W}}_q^{ti} \in \mathbb{R}^{C\times d}$, $\hat{\boldsymbol{W}}_k^{ti} \in \mathbb{R}^{C\times d}$, and $\hat{\boldsymbol{W}}_v^{ti} \in \mathbb{R}^{C\times d}$ are weight matrices for getting query, key, and value, respectively.

The purpose of this process is to compute $\hat{\boldsymbol{Q}}^{t1}$ and $\hat{\boldsymbol{K}}^{t1}$ for the t_1 HSI to obtain the attention map. If a pixel has not changed, the features obtained by aggregating $\hat{\boldsymbol{V}}^{t2}$ from the t_2 HSI should be similar to those obtained by aggregating the features from the t_1 HSI. If a pixel has changed, aligning the features of the t_1 HSI with those of the t_2 HSI will result in discrepancies, as the features extracted from $\hat{\boldsymbol{V}}^{t2}$ will not match $\hat{\boldsymbol{Q}}^{t1}$ and $\hat{\boldsymbol{K}}^{t1}$, thereby amplifying the differences. The same process is applied to the t_2 HSI.

To simplify the representation, we express this as the following equation:

$$\Big(\widetilde{\boldsymbol{x}}^{t1}, \widetilde{\boldsymbol{x}}^{t2}\Big) \triangleq \text{DSCEM}\Big(\boldsymbol{x}^{t1}, \boldsymbol{x}^{t2}\Big). \tag{18}$$

After DSCEM, we can not only focus more on the similar features, but also enhance their difference information synchronously.

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C. Bi-Temporal Image Feature Selection and Fusion Module

To enhance feature representation and reduce redundancy, we design a BFSFM. Using an LSTM-based approach, the BFSFM filters out irrelevant features and fuses complementary information from bi-temporal images. This dynamic fusion process highlights critical features, enabling more accurate identification of changed and unchanged regions and improving the robustness of CD.

Given the intermediate feature $\boldsymbol{A} \in \mathbb{R}^{s\times s\times C}$, we transform it into a matrix $\hat{\boldsymbol{A}} = [\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_{s^2}]^{\mathrm{T}} \in \mathbb{R}^{s^2\times C}$, where $\boldsymbol{a}_k \in \mathbb{R}^C (k = 1, 2, \ldots, s^2)$. For each input $\boldsymbol{a}_k$, the BFSFM retains the essential features and discards the redundant information.

The BFSFM unit is based on the forget gate $\boldsymbol{f}_k$, the input gate $\boldsymbol{i}_k$, the output gate $\boldsymbol{o}_k$, and the memory cell $\boldsymbol{c}_k$. The forget gate $\boldsymbol{f}_k$ allows the module to discard useless information from the memory cell. For the kth step, the forget gate can be described as

$$\boldsymbol{f}_k = \sigma\left(\boldsymbol{W}_{fa}\boldsymbol{a}_k + \boldsymbol{W}_{fh}\boldsymbol{h}_{k-1} + \boldsymbol{b}_f\right) \tag{19}$$

where $\boldsymbol{W}_{fa}$ and $\boldsymbol{W}_{fh}$ denote the coefficient matrices, $\boldsymbol{h}_0$ is defined as $\boldsymbol{a}_1$, $\sigma(\cdot)$ denotes the sigmoid function, and the bias vector associated with the forget gate is denoted as $\boldsymbol{b}_f$.

The input gate, denoted as $\boldsymbol{i}_k$, controls whether new information should be incorporated into the memory cell. The update process for the input gates is carried out by

$$\boldsymbol{i}_k = \sigma(\boldsymbol{W}_{ia}\boldsymbol{a}_k + \boldsymbol{W}_{ih}\boldsymbol{h}_{k-1} + \boldsymbol{b}_i) \tag{20}$$

where the $\boldsymbol{W}_{ia}$ and $\boldsymbol{W}_{ih}$ represent the coefficient matrices for the input gate, while $\boldsymbol{b}_i$ denotes the corresponding bias vector.

The process of updating the memory cell $\boldsymbol{c}_k$ involves selectively removing the old information stored in the previous state $\boldsymbol{c}_{k-1}$ and integrating the new information $\hat{\boldsymbol{c}}_k$

$$\boldsymbol{c}_k = \boldsymbol{i}_k \odot \hat{\boldsymbol{c}}_k + \boldsymbol{f}_k \odot \boldsymbol{c}_{k-1} \tag{21}$$

where $\odot$ represents the pointwise multiplication, $\boldsymbol{c}_0$ is defined as an all-ones vector, and the new candidate cell value, denoted as $\hat{\boldsymbol{c}}_k$, is generated by each cell and can be expressed as follows:

$$\hat{\boldsymbol{c}}_k = \tanh(\boldsymbol{W}_{ca}\boldsymbol{a}_k + \boldsymbol{W}_{ch}\boldsymbol{h}_{k-1}) \tag{22}$$

where $\tanh(\cdot)$ denotes the hyperbolic tangent function, and $\boldsymbol{W}_{ca}$ and $\boldsymbol{W}_{ch}$ are the weight matrices.

The output gate $\boldsymbol{o}_k$ functions as a filter that controls the information being emitted from the memory cell $\boldsymbol{c}_k$. The forward propagation through the output gate can be described by the following expression:

$$\boldsymbol{o}_k = \sigma(\boldsymbol{W}_{oa}\boldsymbol{a}_k + \boldsymbol{W}_{oh}\boldsymbol{h}_{k-1} + \boldsymbol{b}_o) \tag{23}$$

where $\boldsymbol{W}_{oa}$ and $\boldsymbol{W}_{oh}$ represent the coefficient matrices for the output gate $\boldsymbol{o}_k$, with $\boldsymbol{b}_o$ as the corresponding bias vector. The output of the current cell can then be calculated as follows:

$$\boldsymbol{h}_k = \boldsymbol{o}_k \tanh(\boldsymbol{c}_k). \tag{24}$$

To simplify the representation, we express this entire module using the following equation:

$$\boldsymbol{h}_{s^2} \triangleq \mathrm{BFSFM}(\boldsymbol{A}). \tag{25}$$

Algorithm 1 Adaptive Information Weighting and Synchronized Enhancement Network (AIWSEN)

Input: t_1 HSI data $\boldsymbol{X}^{t1}$, t_2 HSI data $\boldsymbol{X}^{t2}$, and ground truth data $\boldsymbol{Y}$

Output: CD map $\hat{y}$

1: Initialize all weights
2: Create pixel patches $\boldsymbol{x}^{t1} \in \mathbb{R}^{s\times s\times C}$ and $\boldsymbol{x}^{t2} \in \mathbb{R}^{s\times s\times C}$ from $\boldsymbol{X}^{t1}$ and $\boldsymbol{X}^{t2}$, and divide them into training and testing sets.
3: **for** $item < I$ **do**
4: Feed the subtracted features into the several AIWAMs for extracting difference feature $\boldsymbol{D}_L$ by Eq. (26).
5: Input bi-temporal features $\boldsymbol{x}^{t1}$ and $\boldsymbol{x}^{t2}$ into several DSCEMs to extract interactive features $\boldsymbol{M}_H$ by Eq. (27) and Eq. (28).
6: Take the $\boldsymbol{D}_L$ and $\boldsymbol{M}_H$ into the BFSFM to obtain the fused feature $\boldsymbol{Z}$ by (29).
7: Input $\boldsymbol{Z}$ into the classifier to obtain the prediction labels.
8: **end for**
9: Obtain CD map $\hat{y}$.

Note: where $item$ represents the index variable for iterating over the training samples, and I denotes the total number of iterations.

After passing through the BFSFM, we can fuse features of bi-temporal images, effectively retaining valid features and discarding redundant information. This process allows for more accurate identification of changed and unchanged regions.

D. Train Setting and Loss Function

For bi-temporal HSI patches $\boldsymbol{x}^{t1} \in \mathbb{R}^{s\times s\times C}$ and $\boldsymbol{x}^{t2} \in \mathbb{R}^{s\times s\times C}$, the differential image $\boldsymbol{D}_0$ is computed as $\boldsymbol{D}_0 = \boldsymbol{x}^{t1} - \boldsymbol{x}^{t2}$, which is sent into the several AIWAMs to learn the difference features. Assume there are L AIWAMs, then the specific formula of the lth AIWAM is as follows:

$$\boldsymbol{D}_l = \mathrm{AIWAM}\Big(\mathrm{BN}\Big(f_{\mathrm{Conv}}^{3\times 3}(\boldsymbol{D}_{l-1})\Big)\Big) \tag{26}$$

where $l = 1, 2, \ldots, L$, $f_{\mathrm{Conv}}^{m\times n}$ denotes a conventional convolution with a kernel size of $m \times n$, and the outputs $\boldsymbol{D}_l$ are of size $(s/2) \times (s/2) \times 2C$. After passing through L layers of AIWAM, we obtain the final difference feature $\boldsymbol{D}_L$.

Then, we send the $\boldsymbol{x}^{t1}$ and $\boldsymbol{x}^{t2}$ into DSCEMs to get the interactive features. There are H DSCEMs in our method. For the hth layer, we have

$$\left(\boldsymbol{x}_h^{t1}, \boldsymbol{x}_h^{t2}\right) = \mathrm{DSCEM}\left(\boldsymbol{x}_{h-1}^{t1}, \boldsymbol{x}_{h-1}^{t2}\right) \tag{27}$$

where $h = 1, 2, \ldots, H$. Particularly, for the input of the first DSCEM, we set $\boldsymbol{x}_0^{t1} = \boldsymbol{x}^{t1}$ and $\boldsymbol{x}_0^{t2} = \boldsymbol{x}^{t2}$. After H DSCEM modules, we get the final results $\boldsymbol{M}_H$

$$\boldsymbol{M}_H = \boldsymbol{x}_H^{t1} - \boldsymbol{x}_H^{t2}. \tag{28}$$

After that, we send the $\boldsymbol{D}_L$ obtained from AIWAMs and the $\boldsymbol{M}_H$ obtained from DSCEM into the feature fusion network.

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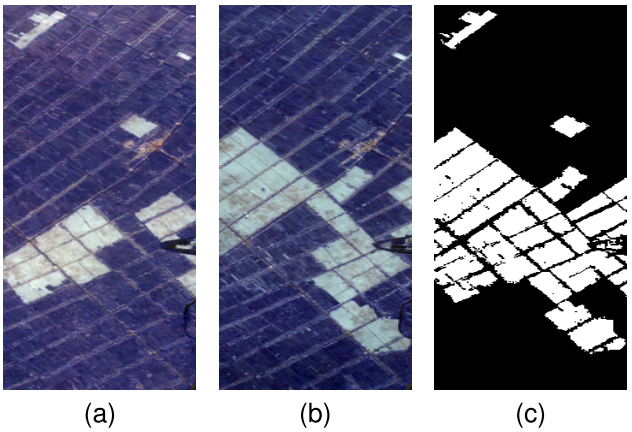


Fig. 2. Farmland dataset. (a) Imagery was acquired on May 3, 2006. (b) Imagery was acquired on April 23, 2007. (c) Ground truth.

The module is composed of two LSTMs, and the fused feature is

$$\boldsymbol{Z} = \text{Concat}(\text{BFSFM}(\boldsymbol{D}_L), \text{BFSFM}(\boldsymbol{M}_H)). \tag{29}$$

Finally, we input the fused feature $\boldsymbol{Z}$ into the classifier to obtain the prediction result $\hat{y}$

$$\hat{y} = \text{Softmax}(\boldsymbol{Z}\widetilde{\boldsymbol{W}}) \tag{30}$$

where $\widetilde{\boldsymbol{W}}$ denotes the weight of the softmax layer.

The cross-entropy loss function is used to train the proposed AIWSEN as follows:

$$\text{Loss} = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right) \tag{31}$$

where n represents the number of training samples, and y_i and $\hat{y}_i$ denote the ground truth and prediction result of the ith sample, respectively. The comprehensive training procedure of the proposed method is shown in Algorithm 1.

III. Experiments

In this section, we first describe the public HSI-CD datasets and introduce five evaluation metrics. Then, we briefly describe comparative methods and the specific experimental details. After that, experiments are conducted to evaluate different methods. Finally, we conduct several ablation studies and analyze the hyperparameters.

A. Datasets

The first dataset, depicted in Fig. 2, represents a farmland area near Yancheng, Jiangsu, China. This dataset was acquired using the Earth Observing-1 (EO-1) Hyperion sensor on May 3rd, 2006, and April 23rd, 2007, respectively. It originally comprised 242 spectral bands ranging from 0.4 to 2.5 μm with a spatial resolution of 30 m. After removing noise and water absorption bands, the dataset was reduced to 155 spectral bands. Each image has a spatial size of 450 $\times$ 140 pixels. The primary area of change within this dataset is the land cover.

The second dataset, illustrated in Fig. 3, captures a river near Jiangsu Province, China. This dataset was acquired using the EO-1 Hyperion sensor on May 3rd, 2013, and December 31st, 2013, respectively. Each image has 463 $\times$ 241 pixels and includes 198 spectral bands. The spatial resolution is 30 m, and the spectral resolution is 10 nm. The primary area of change observed in this dataset is the disappearance of substance in river.

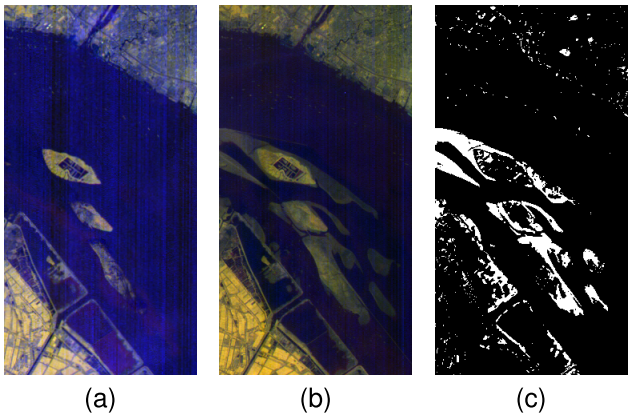


Fig. 3. River dataset. (a) Imagery was acquired on May 2013. (b) Imagery was acquired on December 2013. (c) Ground truth.

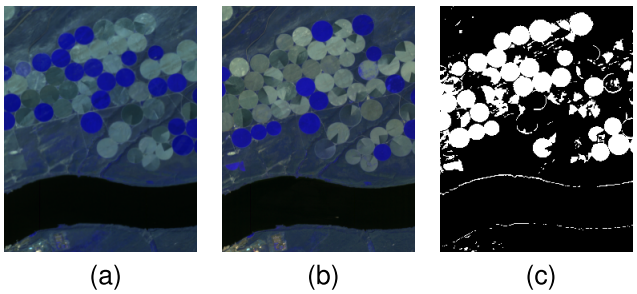


Fig. 4. USA dataset. (a) Imagery acquired in 2004. (b) Imagery acquired in 2007. (c) Ground truth.

The third dataset, displayed in Fig. 4, depicts the area near Hermiston City, CA, USA. This dataset was collected using the Hyperion sensors on May 1st, 2004, and May 8th, 2007, respectively. Each image has 307 $\times$ 241 pixels and includes 154 spectral bands. The primary change observed in this dataset involves farmland cover.

B. Evaluation Metrics

We consider the following CD evaluation metrics, i.e., overall accuracy (OA), Kappa coefficient (κ), precision (P), recall (R), and $F1$ score ($F1$). These metrics are defined as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \tag{32}$$

$$\text{P} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{33}$$

$$\text{R} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{34}$$

$$F_1 = \frac{2\text{P}\times\text{R}}{\text{P}+\text{R}} \tag{35}$$

$$\kappa = \frac{\text{OA}-p_e}{1-p_e} \tag{36}$$

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$$p_e = \frac{(\mathrm{TP}\cdot\mathrm{FN}) + (\mathrm{TP}\cdot\mathrm{FP}) + (\mathrm{TN}\cdot\mathrm{FN}) + (\mathrm{TN}\cdot\mathrm{FP})}{N^2} \quad (37)$$

where true positive (TP) refers to the number of pixels accurately identified as part of a changed region, while true negative (TN) represents the pixels correctly identified as unchanged. False positive (FP) indicates the number of pixels incorrectly classified as changed regions, and false negative (FN) denotes the pixels that were mistakenly classified as unchanged regions. Higher values of the above five metrics signify superior performance. To show the experimental results more clearly, TP, TN, FP, and FN are displayed in white, black, blue, and red on the visualization map.

C. Compared Methods and Experimental Setup

The proposed AIWSEN is compared with the following CD methods, including CVA [7], IR-MAD [8], SVM [9], BIT [24], ML-EDAN [19], CSANet [32], SSTFormer [33], MSDFFN [20], and GTMSiam [21]. CVA is a classical traditional method, which calculates CD results using the Euclidean distance and a threshold. MAD utilizes the canonical correlation analysis to obtain the results. SVM employs a kernel function to maximize the margin between changed and unchanged pixels, thereby achieving classification results. BIT uses a transformer encoder–decoder architecture to model semantic information. ML-EDAN and MSDFFN use multiscale modules to extract subtle features. CSANet and SSTFormer employ cross-temporal interaction symmetric attention and three-layer multihead cross-attention, respectively, to extract spatial, spectral, and temporal features. GTMSiam uses Siamese dual branches to capture multilayer features to get its results.

In the experiments, we randomly select 1% labeled samples for training. For BIT, CSANet, and MSDFFN, the input patch size is 9×9. For ML-EDAN, GTMSiam, and SSTFormer, the input patch size is 5×5. For our network, the input patch size is set as 7×7, and the number of AIWAM and DSCEM layers are set as $L = 4$ and $H = 2$, respectively. For all deep learning methods, the SGD optimizer with a weight decay of 5e-3 is used to optimize the network. The initial learning rate is set to 5e-3 and decayed by a factor of 0.1 every 35 epochs in all experiments. The total number of epochs is 100, and the batch size is set to 64. All methods are trained and tested on an NVIDIA GeForce RTX 2080 GPU with 12 GB of memory using the PyTorch framework.

D. Experimental Results

Table I shows the results of different methods on the three public datasets with 1% labeled training samples. In comparison with the traditional CVA method, deep learning algorithms generally exhibit higher precision and show an improvement in terms of κ. This indicates that most deep learning methods achieve a balance between P and R, resulting in more stable outcomes. Traditional algorithms often misjudge changed regions as unchanged ones, or vice versa, leading to high P but low R on the River dataset, and high R but low P on the USA dataset. Deep-learning-based methods, including ML-EDAN and MSDFFN, generally outperform traditional methods due to the convolutional layers that can automatically extract deep spatial–spectral features. CSANet and SSTFormer use the transformer module to capture global features, resulting in better performance than most traditional methods on all three datasets. On the Farmland dataset, MSDFFN and ML-EDAN achieve better results than other algorithms, suggesting they effectively utilize the multiscale information in HSIs. On the USA dataset, SSTFormer performs better compared to MSDFFN and ML-EDAN, indicating potential overfitting by MSDFFN and ML-EDAN with a labeled sample size of 1%. Compared to all the methods, our proposed AIWSEN achieves the best performance in terms of OA and κ across the three datasets. The OA scores are 97.53%, 96.41%, and 96.42% for the Farmland, River, and USA datasets, respectively. This demonstrates that the proposed AIWAM can effectively capture change information, and the DSCEM can utilize exchange information from bi-temporal images.

Fig. 5 shows the visualization results on the Farmland dataset. From visual observations, the CVA and IR-MAD methods misclassify a large number of unchanged pixels as changed pixels in the red regions and misclassify pixels around the edges of changed areas and small targets. This is because traditional methods cannot simultaneously learn the spatial and spectral information of HSIs well. The BIT, CSANet, and SSTFormer perform better on the unchanged pixels by fully utilizing the spatial and spectral information. However, the edge regions and small targets still show some errors, appearing as blue or red regions. The ML-EDAN, GTMSiam, and MSDFFN achieve better results due to the strength of convolutional kernels, but they still make some misclassifications in detailed regions. Compared to other methods, our proposed AIWSEN shows the best visual performance and presents the fewest red and blue regions. This is because our proposed AIWAM can fully utilize change information to achieve better CD results.

Fig. 6 shows the visualization results on the River dataset. Traditional methods exhibit more misclassified regions and significant "salt and pepper" noise in both changed and unchanged regions. In contrast, deep-learning-based methods, with their advanced neural network architectures, offer improved visual performance by better distinguishing changed and unchanged regions, thereby reducing the occurrence of noise. However, they still misclassify some small changed regions and pixels along the river edges. Our proposed AIWSEN achieves the best visual performance, accurately detecting almost all changed and unchanged regions.

Fig. 7 shows the visualization results on the USA dataset. The CVA and IR-MAD methods obtain the worst results because they misjudge many unchanged pixels as changed ones. This misjudgment leads to significant errors in the detection process, causing areas that should remain unchanged to be incorrectly highlighted as changes. Although the SVM does not misjudge unchanged regions, it misclassifies many changed pixels as unchanged pixels, as shown by the blue regions in Fig. 7(c). The deep learning algorithms achieve better visual performance than traditional algorithms. The changed regions in the USA dataset appear as circular areas with partially connected edges. The deep learning methods may ignore independently scattered regions and some edges

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TABLE I
CD RESULTS ON THE FARMLAND, RIVER, AND USA DATASETS WITH 1% LABELED SAMPLES

Dataset	Methods	OA%	$\kappa\times100$	F1%	P%	R%
Farmland	CVA	95.74	89.83	92.87	90.36	95.51
	IR-MAD	96.69	91.97	94.29	94.46	94.13
	SVM	96.20±0.17	90.84±0.39	93.53±0.27	92.34±0.83	94.77±0.64
	BIT	95.79±0.33	89.94±0.72	92.93±0.47	90.78±1.70	95.22±1.26
	ML-EDAN	96.74±0.14	92.08±0.33	94.37±0.24	94.51±0.91	94.25±1.02
	CSANet	96.43±0.16	91.36±0.36	93.87±0.25	93.61±1.16	94.16±1.05
	SSTFormer	96.26±0.44	91.02±1.04	93.66±0.74	92.38±0.89	94.99±0.82
	GTMSiam	96.46±0.20	91.50±0.43	94.11±0.22	92.31±1.28	95.79±0.87
	MSDFFN	96.75±0.14	92.11±0.37	94.40±0.27	94.28±0.70	94.53±1.09
	AIWSEN	**97.53±0.20**	**94.03±0.45**	**95.77±0.31**	**95.12±1.14**	**96.45±0.92**
River	CVA	95.52	66.08	68.36	88.42	55.72
	IR-MAD	92.92	52.85	56.69	60.49	53.34
	SVM	95.06±0.54	59.98±7.64	62.28±7.63	**92.25±6.71**	48.24±10.72
	BIT	94.91±0.11	62.59±1.99	65.23±1.98	80.25±2.00	55.12±3.78
	ML-EDAN	95.67±0.20	70.35±1.29	72.68±1.20	80.60±2.51	66.25±2.23
	CSANet	95.07±0.18	65.32±2.08	67.94±2.02	78.14±1.77	60.23±3.81
	SSTFormer	95.27±0.43	68.65±2.76	71.22±2.56	75.71±3.33	67.38±3.65
	GTMSiam	94.09±0.41	51.88±8.24	54.61±8.49	82.18±6.69	42.19±9.83
	MSDFFN	95.59±0.28	69.78±0.99	72.15±0.81	80.18±4.39	65.71±1.74
	AIWSEN	**96.41±0.16**	**75.59±1.04**	**77.54±1.00**	84.10±3.41	**72.10±3.38**
USA	CVA	90.76	76.87	82.95	71.02	**99.72**
	IR-MAD	92.39	80.49	85.51	74.85	99.11
	SVM	92.51±1.99	76.99±6.81	81.60±2.74	92.65±5.78	74.98±9.62
	BIT	94.17±0.62	83.27±1.76	87.03±1.35	87.36±1.94	86.72±1.44
	ML-EDAN	94.71±0.44	84.64±1.54	87.70±1.17	92.38±1.95	83.78±2.39
	CSANet	94.35±0.29	83.23±0.93	86.81±0.75	91.56±1.43	82.57±1.76
	SSTFormer	95.49±0.15	86.96±0.33	89.95±0.20	92.18±2.04	89.09±2.41
	GTMSiam	95.12±0.58	85.42±1.92	88.51±1.58	**94.08±1.09**	83.62±2.88
	MSDFFN	94.79±0.62	84.80±1.75	88.14±1.35	90.54±2.27	85.89±1.76
	AIWSEN	**96.42±0.20**	**89.74±0.56**	**92.05±0.44**	92.10±1.21	92.03±1.28

of the circular areas, as shown in Fig. 7(h). Other supervised algorithms, such as SSTFormer and CSANet, show many FN and FP, and some circular connected edges are lost in the CD results. Compared with these methods, our proposed AIWSEN achieves the best visual performance. This is largely due to our network makes full use of the exchange information. By effectively integrating spatial and spectral information, our network can more accurately distinguish changed and unchanged regions, leading to more reliable and comprehensive CD results.

E. Ablation Study

In this section, we conduct ablation experiments on three datasets to evaluate the effectiveness of three key modules: AIWAM, DSCEM, and BFSFM. To achieve this, we design four model variants. The first variant employs only the CNN module to extract difference information, with the extracted features passed to a classifier to generate the final predictions. The second variant exclusively utilizes DSCEM to extract interactive features, followed by difference calculation, pooling, and classification for the final results. The third variant combines AIWAM for extracting difference features and DSCEM for interactive features, where the pooled features are subsequently fused to produce the final output. The fourth and final variant represents the proposed network, which integrates all three modules for comprehensive feature extraction and refinement.

Table II demonstrates that the OA consistently improves as additional modules are integrated. Our proposed model achieves the highest OAs across all datasets, highlighting the effectiveness of each module in enhancing performance. For the Farmland dataset, incorporating AIWAM significantly improves the R-value, indicating that AIWAM effectively captures detailed change information. Furthermore, combining AIWAM with DSCEM enhances the model's performance, demonstrating the complementary roles of AIWAM and DSCEM in leveraging differential and interactive features to boost CD performance. Similarly, the inclusion of BFSFM enables more effective filtering and refinement of essential features, as confirmed by the ablation experiments. The full network, comprising AIWAM, DSCEM, and BFSFM, achieves superior results, underscoring the contributions of

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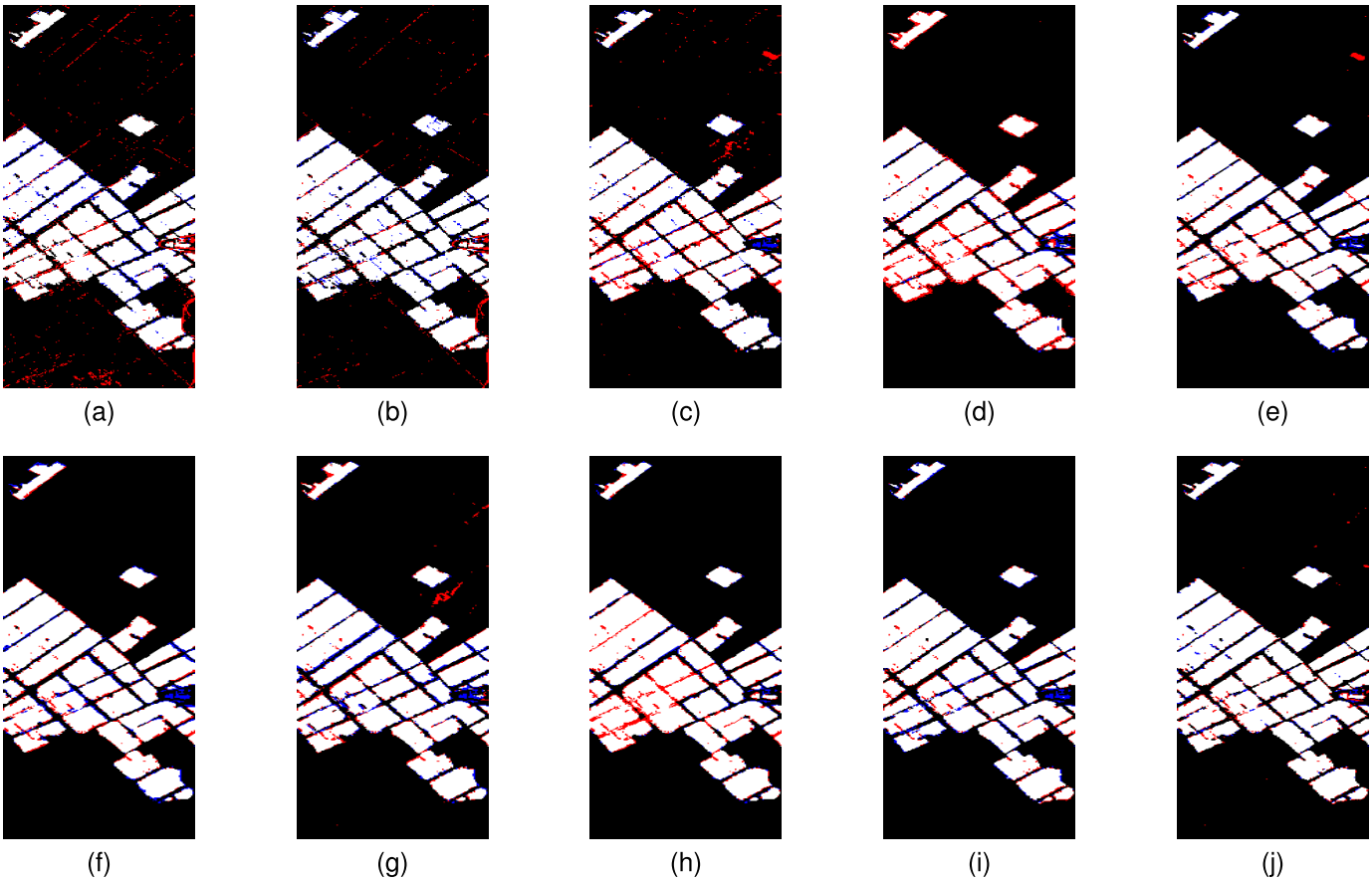


Fig. 5. CD map of all methods on the Farmland dataset with 1% labeled samples. (a) CVA. (b) IR-MAD. (c) SVM. (d) BIT. (e) ML-EDAN. (f) CSANet. (g) SSTFormer. (h) GTMSiam. (i) MSDFFN. (j) AIWSEN. The FN and FP pixels are denoted in red and blue, respectively.

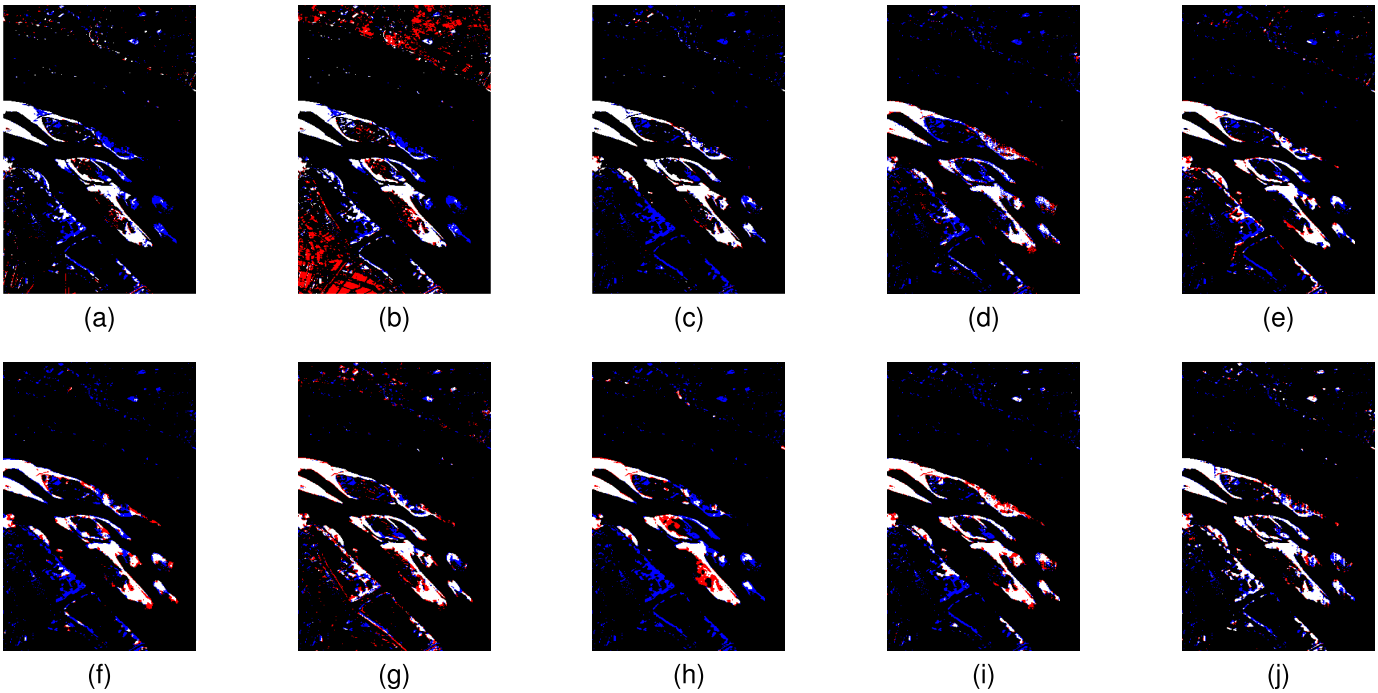


Fig. 6. CD map of all methods on the River dataset with 1% labeled samples. (a) CVA. (b) IR-MAD. (c) SVM. (d) BIT. (e) ML-EDAN. (f) CSANet. (g) SSTFormer. (h) GTMSiam. (i) MSDFFN. (j) AIWSEN. The FN and FP pixels are denoted in red and blue, respectively.

each module. Comparable improvements are observed on the USA dataset. For the River dataset, despite challenges posed by numerous small targets, the proposed AIWSEN still delivers superior performance.

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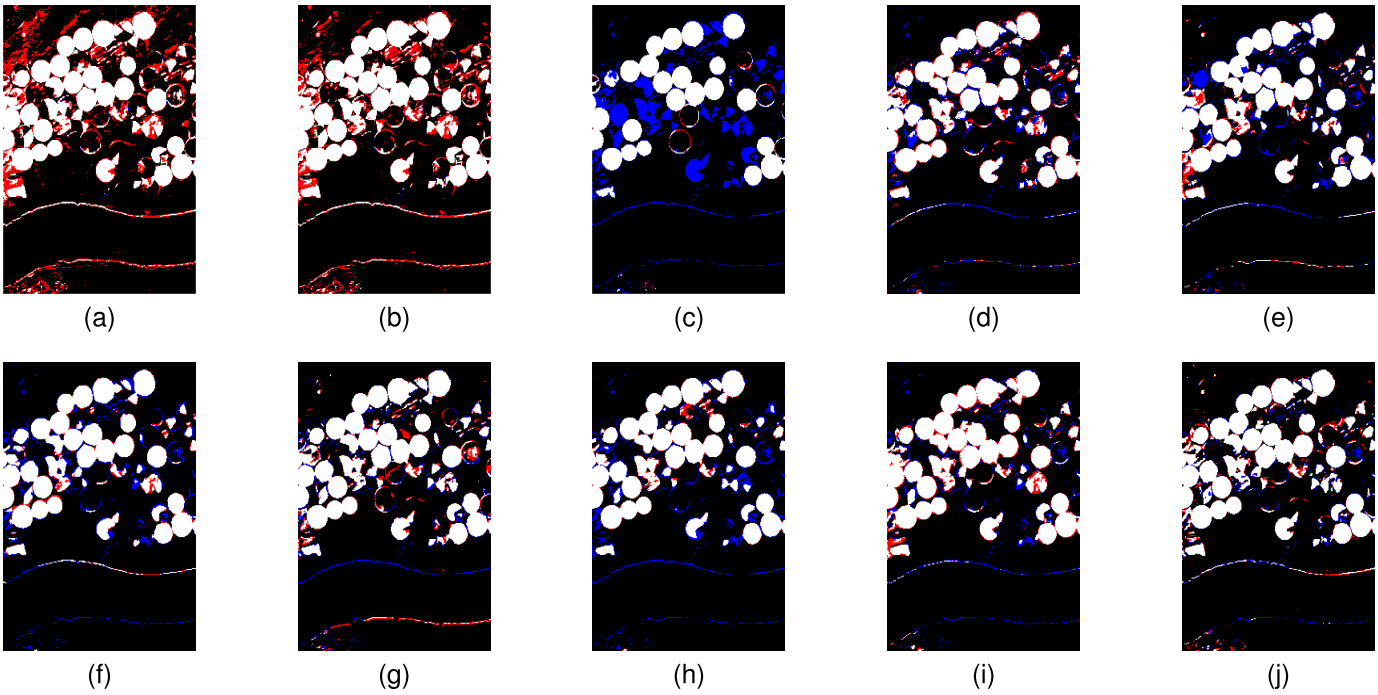


Fig. 7. CD map of all methods on the USA dataset with 1% labeled samples. (a) CVA. (b) IR-MAD. (c) SVM. (d) BIT. (e) ML-EDAN. (f) CSANet. (g) SSTFormer. (h) GTMSiam. (i) MSDFFN. (j) AIWSEN. The FN and FP pixels are denoted in red and blue, respectively.

TABLE II
ABLATION EXPERIMENTS ON THREE DATASETS

Dataset	DSCEM	AIWAM	BFSFM	OA%	$\kappa\times100$	F1%	P%	R%
Farmalnd	×	×	×	91.49±0.59	80.19±1.36	82.98±1.01	86.33±0.86	84.27±1.48
	✓	×	×	93.85±1.13	85.18±2.66	89.53±1.85	88.44±2.52	90.69±1.86
	✓	✓	×	97.32±0.10	93.55±0.25	94.45±0.18	94.25±0.49	**96.67±0.57**
	✓	✓	✓	**97.53±0.20**	**94.03±0.45**	**95.77±0.31**	**95.12±1.14**	96.45±0.92
River	×	×	×	89.10±0.34	45.75±1.20	51.53±1.07	41.99±1.06	66.71±2.02
	✓	×	×	94.05±0.27	62.05±1.63	65.29±1.50	66.29±1.94	64.38±2.16
	✓	✓	×	96.10±0.25	75.19±1.95	77.32±1.82	78.16±1.18	**76.56±3.26**
	✓	✓	✓	**96.38±0.16**	**75.59±1.04**	**77.54±1.00**	**84.10±3.41**	72.10±3.38
USA	×	×	×	88.55±0.54	72.65±2.25	80.95±2.15	81.67±1.54	75.02±3.18
	✓	×	×	93.17±1.21	77.59±1.04	82.31±0.72	88.06±2.77	77.31±0.93
	✓	✓	×	96.05±0.35	88.72±0.99	91.27±0.77	90.91±0.91	91.63±1.01
	✓	✓	✓	**96.42±0.20**	**89.74±0.56**	**92.05±0.44**	**92.10±1.21**	**92.03±1.28**

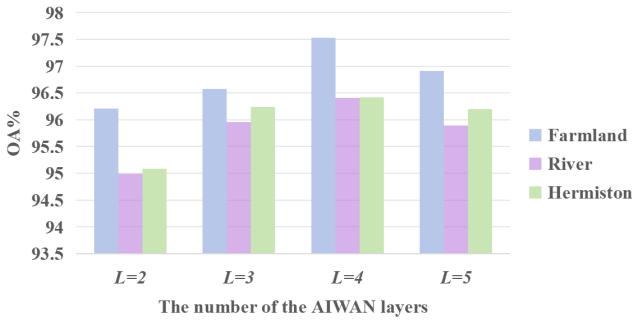


Fig. 8. OA versus the number of AIWAM layers.

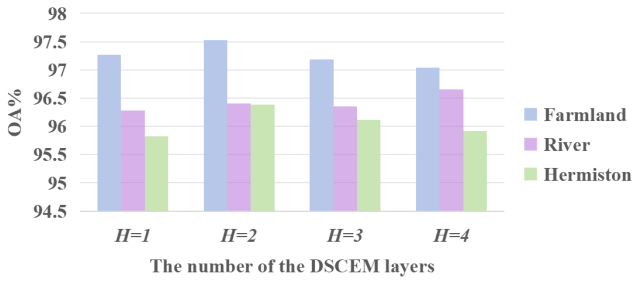


Fig. 9. OA versus the number of DSCEM layers.

F. Discussion

1) Effect of the Number of AIWAM Layers: Fig. 8 shows the OA of AIWSEN versus the number of AIWAM layer L on three datasets, where L varies from 2 to 5. When L is small, the model with too few layers makes it difficult to capture complex spatial and spectral features and hence shows poor results. As the increase of the number of layers, the OA

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generally improves. However, further increasing the number of layers to $L = 5$ leads to a decline in performance. The deeper architecture may introduce the risk of overfitting. As a result, we set $L = 4$ in the experiments.

2) Effect of the Number of DSCEM Layers: Fig. 9 shows the OA of AIWSEN versus the number of DSCEM layer H on three datasets, where H varies from 1 to 4. It can be seen that the model shows the best overall results in the case of $H = 2$. In the experiments, we set $H = 2$ for three datasets consistently.

IV. Conclusion

In this article, we have proposed an AIWSEN for HSI CD. The proposed AIWSEN employs the information entropy to identify change features and extract bi-temporal exchange information, which consists of three main modules: AIWAM, DSCEM, and BFSFM. Specifically, the AIWAM leverages the maximum discrete entropy theorem to detect differences in bi-temporal HSIs. The DSCEM is proposed to extract features by interactively aggregating features from bi-temporal HSIs to enhance difference features. In addition, the BFSFM integrates the extracted features to enhance overall performance. Experimental results on three HSI CD datasets demonstrate the superiority of AIWSEN over existing methods.

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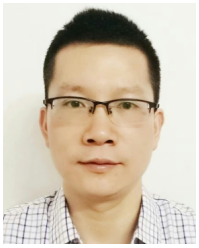
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